

RACER-GT Protocol and Preregistration Template

What to Lock Before Any Data Are Seen

Yi-Hao Lai

Department of Finance, Da-Yeh University

2026-07-28 • version 1.5.0

Abstract

This template is meant to be filled in, timestamped, and deposited *before* collection begins. Its purpose is narrow and specific: to make the retrieval-side researcher degrees of freedom visible. A Google Trends series constructed after the analyst has seen the financial results it will be used to explain is not a measurement; it is a choice. Everything below exists to make that distinction checkable by someone else.

Contents

1	Study identification	3
2	Construct and query specification	3
3	Collection design	3
4	Analysis settings, locked in advance	4
5	Acceptance thresholds	4
6	Pilot separation	4
7	Downstream use	5

8 Deviations

5

1 Study identification

- Title, authors, affiliations, contact.
- Deposit location and timestamp (OSF, AEA RCT Registry, institutional repository, or a signed Zenodo deposit).
- RACER-GT version used.
- Protocol hash produced by `racergt design`. Record it here. Any later change to a locked setting changes this hash, which is what makes drift detectable rather than deniable.

2 Construct and query specification

State the economic construct first and the query second, in that order, because the reverse order invites choosing whichever query gives the desired result.

- The construct the series is intended to measure.
- Why this keyword or Topic measures it, and what else it plausibly picks up.
- `keyword`, `topic_or_term`, `geo`, `category`, `search_property`, `language`.
- Justification for the category choice. Restricting to Finance changes the construct; so does not restricting. Say which you chose and why.
- Historical start and end dates, and the baseline window for normalization.

3 Collection design

- Collection-day offsets, streams, technical replicates, time slots.
- What each stream holds fixed: device, browser profile, cookie state, account state, network route, IP.
- Declaration: an estimated stream effect will not be reported as an IP effect unless a crossover factorial design varying IP alone was run.
- Chunk window length, step, and minimum overlap.
- The collection route (official API, manual export, institution-approved client) and its compliance basis.

4 Analysis settings, locked in advance

Every field of the configuration is part of the protocol. State at minimum: calibration threshold $c_{\min}$, Huber constant, reference-chunk strategy, normalization mode, duplicate thresholds, covariance estimator, weight cap, G-study transformations, benchmark mode and weights.

5 Acceptance thresholds

Fill in every threshold *before* collection, and commit to reporting the resulting status whatever it is.

Rule	What it protects against	Your value
min_unique_pulls	a batch that is mostly duplicates dependence counted as replication	
min_spectral_effective_pulls		
max_zero_share	daily frequency unsupportable for a rare term	
min_level_generalizability	levels that do not reproduce	
min_innovation_generalizability	daily changes that do not reproduce	
min_detection_kappa	unstable zero/non-zero pattern one dependence cluster dominating	
max_component_share		
max_convergence_mae_100	consensus not yet converged	
max_benchmark_standardized_rmse	daily and weekly series disagreeing	

6 Pilot separation

State explicitly whether an initial monitoring window will be used, and commit to the following rule: if monitoring leads to *any* change in thresholds, stitching rules, query, code, or the decision tree, the monitored data become pilot data and the confirmatory batch restarts from a new Day 0. Without this commitment the same data design and validate the acceptance rule, which produces acceptance bias that no amount of subsequent care removes.

7 Downstream use

- The financial model the index will enter, specified now.
- Whether it enters in levels or differences. If in differences, the innovation generalizability threshold is the binding one, not the level one.
- How measurement error will be handled: reliability-corrected OLS, SIMEX, or both.
- Declaration: GT processing rules are locked before any return, volatility, spread, or VaR result is examined.

8 Deviations

Reserve a section, to be completed after the fact, recording every departure from this protocol, when it was decided, and why. A protocol with an honest deviations section is more credible than one with none.